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BCSE497J – PROJECT I LITERATURE SURVEY REPORT

Research Paper Relationship Analyzer:

**A Survey on Knowledge Graph Construction, Cross-Document Relation Extraction,
Contradiction Detection and LLM-Assisted Research Gap Discovery**

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Abstract

The exponential growth of scientific literature has made it increasingly difficult for researchers to manually track relationships, overlaps, and contradictions across published work. This literature survey examines eighteen research contributions spanning knowledge graph construction, information extraction, cross-document relation extraction, semantic similarity, contradiction detection, and automated research-gap discovery. These themes together form the technical foundation for the proposed mini-project, a Research Paper Relationship Analyzer, which aims to automatically ingest research papers, extract entities and relationships, construct a dynamic knowledge graph, detect contradictions between papers, and surface unexplored research gaps using Large Language Model (LLM)-assisted reasoning. The survey synthesizes the objectives, methodologies, models, key findings, and limitations of the reviewed works, presents a comparative analysis across the studies, and consolidates the recurring research gaps that motivate the proposed system. The review shows that while individual components such as knowledge graph construction, relation extraction, and topic-modelling-based gap discovery are well studied, an end-to-end, explainable, and LLM-integrated pipeline capable of reasoning across multiple full-length research papers — particularly for contradiction detection and research-gap identification — remains largely unexplored, which is precisely the space the proposed project intends to address.

1. Introduction

The volume of scientific publications produced every year has grown far beyond what any individual researcher can manually read, compare, and synthesize. Identifying how papers relate to one another — through shared methods, contradicting results, or complementary findings — is essential for literature review, meta-analysis, and identifying promising directions for future research. Traditional literature review is manual, slow, and prone to bias, while existing automated tools typically focus on a single sub-task such as named entity recognition, relation extraction, or topic modelling, without connecting these sub-tasks into a coherent, reasoning-capable pipeline.

The proposed mini-project, titled the Research Paper Relationship Analyzer, aims to build a system that ingests a corpus of research papers, extracts key entities (methods, datasets, models, results), constructs and evolves a knowledge graph representing relationships between papers, identifies contradictory claims across the literature, and highlights unexplored research gaps — all assisted by modern Large Language Models. To ground this project in existing research, this survey reviews eighteen papers spanning six interconnected themes: (i) automatic knowledge graph construction, (ii) LLM-based information extraction and named entity recognition, (iii) cross-document relation extraction, (iv) scientific claim verification and contradiction detection, (v) automated research-gap discovery, and (vi) semantic similarity and reasoning. Section 2 presents the thematic literature survey. Section 3 provides a comparative analysis across all reviewed works. Section 4 consolidates the research gaps identified across the literature. Section 5 positions the proposed system relative to existing work, and Section 6 concludes the survey.

2. Literature Survey

2.1 Automatic Knowledge Graph Construction

Knowledge graphs (KGs) provide a structured way to represent entities and their relationships, and are central to the proposed system's design. Five of the reviewed works directly address how such graphs can be built, enriched, and applied.

Zhong et al. (2022), "A Comprehensive Survey on Automatic Knowledge Graph Construction," ACM Computing Surveys — reviews over 300 methods across the full KG lifecycle, organizing the field into a three-stage pipeline of knowledge acquisition, refinement, and evolution, and covering entity recognition, relation extraction, entity linking, coreference resolution, KG completion, fusion, and temporal/dynamic KG construction. While extremely comprehensive, the survey does not propose a unified implementation, and the authors flag cross-document reasoning, dynamic KG updating, and integration with LLM reasoning as open problems — all of which are directly relevant to the proposed project's need to build and continuously update a graph from an evolving paper corpus.

Ji et al. (2021), "A Survey on Knowledge Graphs: Representation, Acquisition and Applications," IEEE TNNLS — complements the above by focusing on representation learning (e.g., TransE, TransR), acquisition, temporal KGs, and knowledge-aware applications using architectures such as GNNs, Transformers, and reinforcement learning. It offers a strong theoretical foundation for graph representation but does not address LLM integration or contradiction detection, leaving explainable reasoning and efficient graph representation as open research gaps.

Di Iorio et al. (2014), "Analysing and Discovering Semantic Relations in Scholarly Data" — an earlier but foundational Linked-Open-Data study, proposes the Semantic Lancet Project, which enriches scholarly publications with RDF triples, SPAR ontologies, and citation-context extraction to expose machine-readable relationships that are otherwise buried in unstructured text. Although rich in semantic metadata, the approach depends heavily on manual ontology engineering and predates LLM-based automation, motivating the need for automatic, LLM-assisted semantic enrichment of scholarly relationships.

Milosevic and Thielemann (2022), "Relationship Extraction for Knowledge Graph Creation from Biomedical Literature" — compares rule-based, classical machine-learning, and Transformer-based (T5) methods for extracting biomedical relationships, with the fine-tuned T5 model achieving an F1-score of 0.88 and clearly outperforming rule-based and classical baselines. The pipeline (NER → entity normalization → relation extraction → KG construction) mirrors the architecture required for the proposed system, though it remains restricted to the biomedical domain and curated ontologies.

Yu et al. (2020), "A Relationship Extraction Method for Domain Knowledge Graph Construction, World Wide Web (Springer)" — proposes a convolutional residual network combined with co-occurrence and semantic analysis to extract hierarchical and non-hierarchical relations from heterogeneous food-domain data, storing the resulting graph in Neo4j. The method integrates well across structured, semi-structured, and unstructured sources but is domain-specific, underscoring the need for a generalizable, cross-domain relation extraction approach.

Knowledge Graph Enhanced Large Language Models (OpenReview) — examines how retrieval-augmented reasoning and graph-based knowledge injection can reduce hallucination and improve factual consistency in LLM outputs by combining structured KG retrieval with generative reasoning. This directly informs how the proposed system can use its constructed knowledge graph to ground LLM-based reasoning about paper relationships, although the paper notes that efficient real-time KG updating and scalable retrieval remain unresolved challenges.

2.2 LLM-Based Information Extraction and Named Entity Recognition

Since the proposed system must first extract structured information (entities, methods, results) from raw research papers, four reviewed works examine how modern LLMs perform information extraction and named entity recognition, often in zero- or few-shot settings.

Xu et al. (2024), "Large Language Models for Generative Information Extraction: A Survey," *Frontiers of Computer Science* — provides the first comprehensive survey of generative IE, unifying NER, relation extraction, and event extraction under a single LLM-based framework using prompting, fine-tuning, instruction tuning, and few/zero-shot learning with models such as GPT-4, LLaMA, and Flan-T5. The survey highlights hallucination, high computational cost, and inconsistent structured outputs as persistent limitations, and calls for tighter integration between generative IE and knowledge graph construction — a gap the proposed project directly targets.

Sciannameo et al. (2024), "Information Extraction from Medical Case Reports Using OpenAI InstructGPT," *Computer Methods and Programs in Biomedicine* — demonstrates that zero-shot prompting with GPT-3/InstructGPT can extract structured fields directly from 208 medical case-report PDFs with high accuracy (94% for sex, 82% for age, 94% for injury object, and 89% for body part, improving to 97% after excluding incomplete cases), without any task-specific training or preprocessing. This confirms the feasibility of PDF-to-structured-data extraction that the proposed system's document-ingestion module relies on, though the approach is domain-specific and sensitive to prompt quality and document structure.

Zhou et al. (2023), "PromptNER: Prompting for Named Entity Recognition" (arXiv:2302.05019) — reframes NER as a generative task through prompt engineering and instruction tuning, achieving competitive or superior performance to supervised NER baselines with limited annotated data and strong zero-/few-shot capability. Its main limitation is that extracted entities are not connected to any downstream knowledge graph or relation-extraction stage, which is a gap the proposed pipeline aims to close by feeding extracted entities directly into graph construction.

Baldini Soares et al. (2020), "Matching the Blanks: Distributional Similarity for Relation Learning" (arXiv:2002.00388) — introduces a self-supervised "matching-the-blanks" objective that masks entity mentions to learn transferable, contextual relation representations with Transformer models, reducing reliance on manually labelled relation-extraction datasets. While effective for representation learning, the method does not address document-level or cross-document relation extraction or downstream KG population, both of which are essential for the proposed system.

2.3 Cross-Document Relation Extraction

Because the proposed system must relate findings across multiple papers rather than within a single document, two works specifically addressing cross-document relation extraction are particularly relevant.

Wang et al. (2022), "Entity-Centered Cross-Document Relation Extraction," EMNLP 2022 — improves cross-document relation extraction on the CodRED benchmark using entity-based document filtering, bridge-entity discovery, and cross-path entity-relation attention, achieving over a 10% F1 improvement compared to prior state-of-the-art methods. This entity-centered filtering approach is one of the most directly applicable techniques for the proposed system's cross-paper relationship discovery, though it remains computationally expensive and dependent on the availability of bridge entities linking documents.

Yao, Riedel, and McCallum (2010), "Collective Cross-Document Relation Extraction Without Labelled Data," EMNLP 2010 — an earlier but foundational work, uses distant supervision with a factor-graph model and Gibbs sampling to perform label-free, collective cross-document relation extraction, reporting 13–15% precision improvements over pipeline and isolated baselines. It establishes the theoretical basis for scalable, annotation-free cross-document extraction, but relies on older statistical models with limited semantic understanding compared to modern Transformer- or LLM-based approaches.

2.4 Scientific Claim Verification and Contradiction Detection

Detecting contradictions between papers is a core objective of the proposed system, and two reviewed works tackle this problem directly, albeit within specific domains.

Fang, Fu, and Torvik (2025), "Automatic Scientific Claims Verification with Pruned Evidence Graph," ICLR 2025 Agentic AI Workshop — proposes a GraphRAG-based pipeline that retrieves evidence, constructs and prunes an evidence graph, and performs multi-hop reasoning to predict claim stance on the SciFact dataset. The pruned evidence graph improves rationale selection and supports multi-hop scientific reasoning, but the work is early-stage, sensitive to evidence quality, and computationally demanding — while still leaving cross-paper contradiction reasoning and lightweight GraphRAG as open challenges directly relevant to the proposed system.

Makhervaks, Gillis, and Radinsky (2023), "Clinical Contradiction Detection," EMNLP 2023 — detects contradictions in medical literature using ontology-based distant supervision and fine-tuned NLI models (DeBERTa, BioELECTRA) trained on PubMed, SNOMED, and MedNLI data, along with a newly generated contradiction dataset. The approach achieves strong domain-specific accuracy but is tightly coupled to clinical ontologies and does not generalize outside healthcare, highlighting the need for a domain-independent, explainable contradiction-detection method for general scientific literature.

2.5 Automated Research-Gap Discovery

Automatically surfacing unexplored research directions is a key deliverable of the proposed system. Two reviewed works approach this from complementary angles — large-scale topic modelling and organizational knowledge-network analysis.

Abd-alrazaq et al. (2024), "Machine Learning-Based Approach for Identifying Research Gaps: COVID-19 as a Case Study," JMIR Formative Research — applies a BERTopic pipeline (BERT embeddings, c-TF-IDF, UMAP, HDBSCAN) to 33,206 abstracts from the 1.1-million-paper CORD-19 dataset to automatically cluster topics and surface under-explored areas. The approach is scalable and reduces manual review effort, but is limited to abstract-level text and pure topic modelling without semantic reasoning or uncertainty quantification — the authors explicitly call for full-paper analysis and combining topic modelling with semantic reasoning, which the proposed system aims to deliver.

Qi et al. (2020), "Research on Knowledge Gap Identification Method in Innovative Organizations under the 'Internet+' Environment," Information (MDPI) — combines Word2Vec-based topic representation, TF-IDF demand identification, and eigenvector-centrality network analysis with fuzzy evaluation to quantitatively identify organizational knowledge gaps, outperforming subjective approaches such as SWOT analysis. While effective for organizational contexts, the method offers limited semantic reasoning beyond keyword-level networks and has no LLM or knowledge-graph integration, both of which the proposed system incorporates.

2.6 Semantic Similarity and Semantic Reasoning

Measuring semantic similarity underlies several components of the proposed system, including paper clustering, duplicate detection, contradiction detection, and knowledge-graph edge creation. Two reviewed works address this from a methodological and an architectural perspective respectively.

Chandrasekaran and Mago (2021), "Evolution of Semantic Similarity — A Survey," ACM Computing Surveys — traces the evolution of semantic similarity measurement from knowledge-based and corpus-based methods (e.g., WordNet, TF-IDF, LSA) through deep-learning approaches (Word2Vec, GloVe, LSTM, Transformer, BERT), benchmarking them across datasets such as STS2012–2017 and SICK. The survey is comprehensive but does not focus on scientific-literature analysis or knowledge-graph integration, leaving cross-document semantic similarity and explainable similarity measures as open problems relevant to the proposed system.

Okafor and Bello (2024), "Designing a Semantic Analysis Framework for Intelligent Learning by Reading Systems Using Advanced Text Comprehension Techniques" — proposes a hybrid symbolic–neural architecture that combines Transformer embeddings, knowledge graphs, and predicate logic in a multi-stage inference pipeline to improve reading comprehension and explainability. The framework demonstrates strong reasoning and interpretability but remains a computationally intensive prototype without large-scale validation, pointing to the need for scalable semantic reasoning over long, full-length scientific documents.

2.7 Common Observations Across the Reviewed Literature

Across all eighteen papers, several recurring patterns emerge. Knowledge graphs are widely and successfully used to organize scientific and domain knowledge, but end-to-end automated construction pipelines remain incomplete. LLM integration is growing rapidly but is still largely limited to individual sub-tasks (NER, IE, claim verification) rather than being woven into a unified reasoning pipeline. Contradiction detection is almost exclusively studied within narrow domains such as clinical or biomedical NLP, with no general-purpose, cross-domain solution. Research-gap discovery methods rely primarily on topic modelling over abstracts rather than deep semantic reasoning over full papers. Cross-document and cross-paper reasoning is addressed by only a handful of works and remains computationally expensive. Finally, explainability and dynamic/incremental knowledge updating are consistently flagged as unresolved challenges across nearly every reviewed study.

3. Comparative Analysis

Table 1 consolidates all eighteen reviewed works across their focus area, core method, principal contribution, and main limitation, enabling a side-by-side comparison of how each study relates to the components required by the proposed Research Paper Relationship Analyzer.

Table 1. Comparative summary of the reviewed literature

#	Authors (Year)	Focus Area	Core Method / Model	Key Result / Contribution
1	Zhong et al. (2022)	KG Construction (Survey)	3-stage KG pipeline; BERT, GNN	Reviews 300+ KG construction methods
2	Ji et al. (2021)	KG Representation & Acquisition	TransE/TransR, GNN, Transformer	New taxonomy of KG learning
3	Fang et al. (2025)	Scientific Claim Verification	GraphRAG, Pruned Evidence Graph	Improved rationale selection & stance prediction
4	Makhervaks et al. (2023)	Clinical Contradiction Detection	DeBERTa, BioELECTRA + NLI	New clinical contradiction dataset
5	Abd-alrazaq et al. (2024)	Research Gap Discovery	BERTopic, UMAP, HDBSCAN	Automated topic-based gap discovery on 33K abstracts
6	Chandrasekaran & Mago (2021)	Semantic Similarity (Survey)	TF-IDF to BERT/Transformer	Complete taxonomy of similarity methods
7	Wang et al. (2022)	Cross-document Relation Extraction	Entity-centered attention, bridge entities	>10% F1 improvement on CodRED
8	Yao et al. (2010)	Cross-document Relation Extraction	Distant supervision, Factor Graph	13–15% precision gain, label-free
9	Xu et al. (2024)	LLM-based Generative IE (Survey)	GPT-4, LLaMA, Flan-T5, prompting	First unified survey of generative IE
10	Sciannameo et al. (2024)	Medical Information Extraction	GPT-3 / InstructGPT, zero-shot	94% accuracy (sex/injury) from case-report PDFs
11	Qi et al. (2020)	Knowledge Gap Identification	Word2Vec, TF-IDF, Eigenvector centrality	Quantitative gap ID beats SWOT/Venn
12	Okafor & Bello (2024)	Semantic Analysis Framework	Symbolic + Neural hybrid, KG	Improved explainability & reasoning
13	Di Iorio et al. (2014)	Scholarly Semantic Relations	RDF, SPAR ontologies, SPARQL	Semantic Lancet triplet store for citations
14	Milosevic & Thielemann (2022)	Biomedical KG Relation Extraction	T5, rule-based, classical ML	T5 model F1 = 0.88
15	Yu et al. (2020)	Domain KG Construction	CNN + Residual Network, Neo4j	Multi-source hierarchical relation extraction

#	Authors (Year)	Focus Area	Core Method / Model	Key Result / Contribution
16	Zhou et al. (2023) – PromptNER	Prompt-based NER	LLM prompting / instruction tuning	Strong zero-/few-shot NER
17	Baldini Soares et al. (2020)	Relation Representation Learning	“Matching the Blanks” self-supervision	Transferable relation embeddings
18	KG Enhanced LLMs (OpenReview)	KG-Augmented LLM Reasoning	Retrieval-augmented generation + KG injection	Improved factuality, reduced hallucination

Table 2. Mapping of reviewed papers to proposed project components

Project Component	Supporting Papers (#)
Knowledge Graph Construction	1, 2, 13, 14, 15, 18
Information Extraction / NER	9, 10, 16, 17
Cross-document Relation Extraction	7, 8
Contradiction Detection / Claim Verification	3, 4
Research Gap Discovery	5, 11
Semantic Similarity & Reasoning	6, 12
LLM-Augmented Reasoning	9, 10, 16, 17, 18

4. Consolidated Research Gaps

Synthesizing the research-gap statements from all eighteen reviewed works reveals a consistent set of unresolved challenges that directly motivate the proposed system:

- Lack of end-to-end automation: most pipelines automate individual stages (extraction, KG construction, or gap discovery) but not the full workflow from raw papers to actionable insights.
- Weak integration between Information Extraction and Knowledge Graph construction, with extracted entities rarely flowing automatically into a live, queryable graph.
- Limited cross-document and cross-paper reasoning, particularly multi-hop reasoning that spans more than two documents at a time.
- Contradiction detection remains domain-specific (chiefly clinical/biomedical) with no general-purpose, explainable, cross-domain solution.
- Research-gap discovery relies mainly on abstract-level topic modelling rather than full-paper semantic reasoning, and rarely quantifies uncertainty in the discovered gaps.
- LLM hallucination and inconsistent structured outputs continue to undermine reliability in generative information-extraction pipelines.
- Dynamic and incremental knowledge graph updating is inadequately supported, limiting systems' ability to stay current as new papers are published.
- Explainability of both the extraction process and the final relationships/contradictions/gaps identified remains a widely acknowledged open problem.

- Prompt sensitivity and lack of standardization in LLM-based extraction reduce reproducibility across studies and domains.

5. Positioning of the Proposed System

The proposed Research Paper Relationship Analyzer is positioned to directly address the gaps identified above by combining, into a single pipeline, the strongest elements observed across the reviewed literature: LLM-based zero-/few-shot information extraction (informed by PromptNER, Matching the Blanks, and the Generative IE survey) for entity and relation extraction from full papers rather than abstracts; entity-centered cross-document relation extraction techniques (following Wang et al. and Yao et al.) to connect findings across multiple papers; automatic knowledge graph construction and evolution (building on Zhong et al., Ji et al., and the biomedical/domain-specific KG pipelines of Milosevic & Thielemann and Yu et al.) to maintain a continuously updated structured representation of the literature; pruned evidence-graph-based contradiction detection (extending Fang et al. and Makhervaks et al.) generalized beyond the clinical domain; and semantic-similarity-driven, full-paper research-gap discovery (extending the BERTopic-based approach of Abd-alrazaq et al. and the network-based method of Qi et al.) enhanced with LLM reasoning rather than topic clustering alone. By unifying these previously siloed capabilities and grounding LLM reasoning in a dynamically constructed knowledge graph (as suggested by the KG-Enhanced-LLM line of work), the proposed system aims to reduce hallucination, improve explainability, and deliver the end-to-end automation that the current literature consistently identifies as missing.

6. Conclusion

This survey reviewed eighteen research works spanning knowledge graph construction, LLM-based information extraction, cross-document relation extraction, contradiction detection, research-gap discovery, and semantic similarity, all of which collectively inform the design of the proposed Research Paper Relationship Analyzer. While each theme has produced mature and effective techniques individually, the literature consistently shows that no existing system unifies these capabilities into a single, explainable, end-to-end pipeline capable of reasoning across multiple full-length research papers. This gap between well-developed component technologies and the absence of an integrated, LLM-assisted system is precisely the space that the proposed mini-project intends to occupy, and the findings of this survey will directly guide the architectural and methodological choices made in the subsequent design and implementation phases of the project.

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